The Macroeconomic Limits of AI Compute

Can the build-out earn its carry — and what growth does it take?

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A community estimate: every number in an open register, every chart regenerated from it.
Paper v26.5, 1 October 2026.

This deck is a rendering of the paper. Where they differ, the paper and the register are right.

What we are trying to do

Replace assertion with arithmetic that anyone can check and improve.

The question. Five companies will spend about $780B this year on AI data centers. Will the revenue arrive to pay for it — and how fast must it grow?

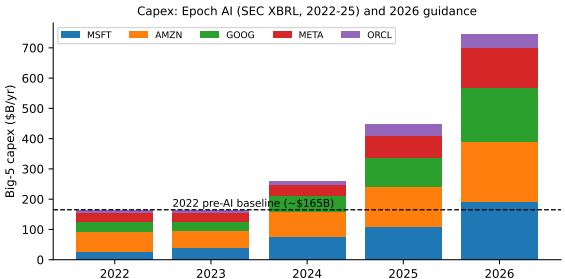
The method, in five steps.

1. Build the *carry* of the installed base, cohort by cohort.
2. Estimate demand twice, independently: top-down (observed 2026 revenue grown at g) and bottom-up (fifteen bounded sector chains).
3. Ask what fraction of a customer's dollar reaches the data-center owner.
4. Propagate every range honestly; report a distribution.
5. Put every input in a public register; let the community move it.

Why Fermi. Fermi got the Trinity yield to within a factor of two with one loose factor and knowledge of which one it was. Here every factor is bounded to tens of percent; "Fermi" names the decomposition and the propagation, not the precision.

Why it matters: the scale of the thing

$780B a year is 1.5–2.5% of US GDP — the peaks of the railway and telecom booms, each of which ended in impairment without ending the technology.



Big-5 capex	$B
2022 (pre-AI baseline)	162
2023	161
2024	260
2025	448
2026 (guidance)	∼780
2027 (street consensus)	∼1,050
2028 (street consensus)	∼1,150

World GDP ∼$115T; US ∼$30T. The question is who carries the write-down and when, not whether AI is affordable.

Source: Epoch AI (SEC XBRL) 2022–25; company guidance Q2 2026; S&P Capital IQ consensus via a16z, Sept 2026

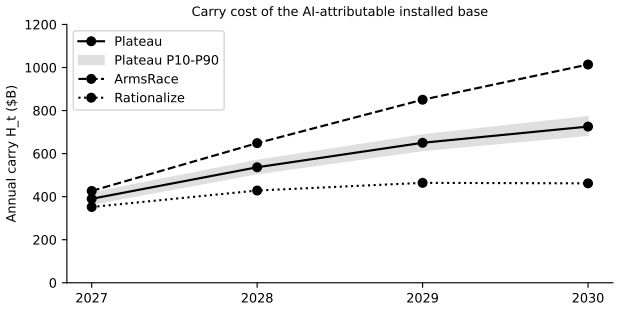
The observables the model must reconcile with

Audited filings first; company statements second; press third — and marked as such.

Quantity	Value	Provenance
Nvidia data-center revenue FY26; Q1 FY27 run-rate	$194B; ∼$300B	8-K (audited)
Big-4 2026 capex guidance	$725–745B	Q2 calls (company statements)
Google Cloud Q2 2026; backlog	+82%; $514B	Alphabet release
Azure FY26 Q4; M365 commercial seats	+43%; +6–7%/yr	Microsoft 8-K metrics
Microsoft 365 Copilot paid seats	>30M (Jul.)	Earnings call (paid ≠ used)
Anthropic run-rate Dec-25 / Jul-26; expected exit-26	$9B / $65B; >$100B	Company; Bloomberg (press)
OpenAI Q1 / Q2 2026 booked; Aug run-rate	$5.7B / $6.7B; $40B	Sacra, press (unaudited)
Oracle RPO Q1 FY27; share from OpenAI; cloud backlogs (3 largest)	$664B; >50%; ∼$1.7T	Earnings call; filings
H100 rental, Sept 2026	$2.63/GPU-hr, rising	Silicon Data
US DC projects blocked/delayed, Q1 2026	$130B	Data Center Watch
Calendar-2026 AI revenue, all channels (derived)	$187B (171–202)	Seven-component decomposition

The carry is a stock, not a flow — and it ramps

Net of the pre-AI baseline, the AI stock is ∼$1.1T entering 2027 and ∼$2.2T by 2029. Its carry: ∼$390B (2027), $535B (2028), $725B (2030) if capex merely plateaus.



Carry H_t = depreciation + facility opex + return on net book, summed over every cohort still alive.

Silicon: effective life 4.2 yr (training 3.5, inference 5, harmonic mean). Shell: 20 yr.

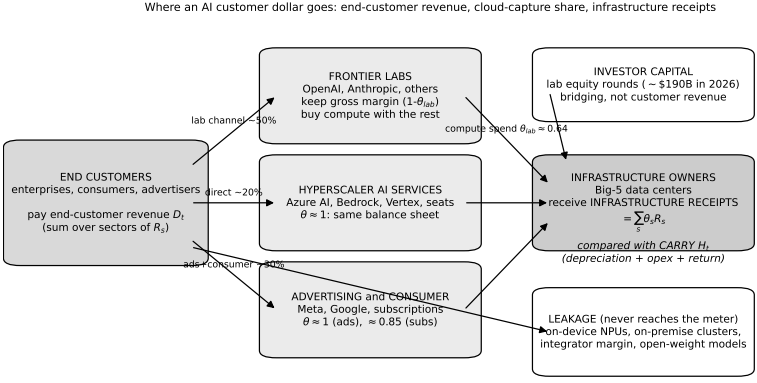
2028 plateau, at the modes: depreciation ∼$280B, opex ∼$130B, return ∼$125B.

A 35% capex cut flattens the carry near $460B — the 2025–26 cohorts are on the books whatever happens next.

Source: paper Section 4, Eq. 2; register entries supply.*

Where a customer's dollar goes

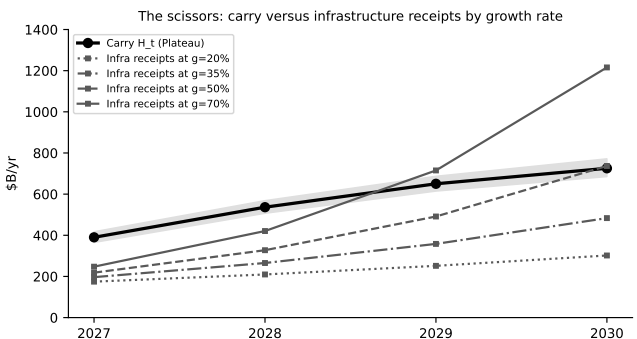
End-customer revenue is not infrastructure receipts. Labs keep a margin; devices, premises and integrators take a share; what is left must cover the carry.



Source: paper Section 6; cloud-capture share $\theta \approx 0.82$ overall, $\theta_{\text{lab}} \approx 0.64$

Top-down: the scissors

Carry on one blade, receipts on the other. Whether and when they cross depends on one number: sustained growth g.



Base: calendar-2026 AI revenue $187B, seven sourced components.

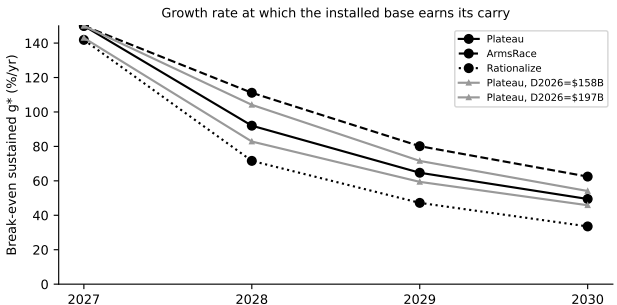
At 20%/yr: never crosses.
At 35%: crosses after 2030.
At 50%: crosses in 2030.
At 70%: crosses in 2029.

No rate crosses in 2027 or 2028: those years are a sunk deficit of $100–250B under every path.

Source: paper Fig. 9; register demand.base_2026, demand.theta

Break-even growth, by year and by capex path

Plateau: ∼48%/yr sustained through 2030. Street-consensus (arms race): ∼62%. A 35% cut: ∼33%.



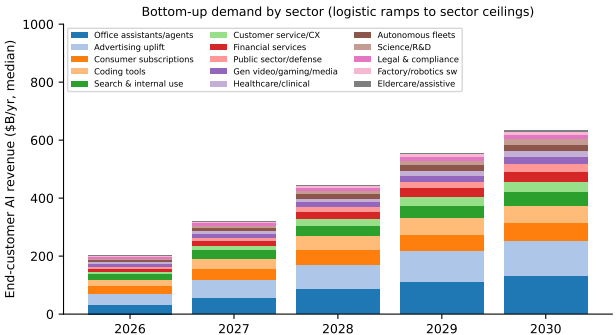
	2028	2029	2030
Plateau	92%	65%	48%
Arms race	111%	80%	62%
Rationalize	72%	47%	33%

Longer silicon life (8 yr) does not help by 2030: cohorts stay on the books longer. A 2030 break-even at 48% means $880B of AI revenue — larger than today's entire public cloud.

Source: paper Table 8

Bottom-up: fifteen sector chains, no growth rate assumed

Each sector has a 2026 base, a 2030 ceiling set by a physical or budgetary constraint, a ramp, and a cloud-capture share. Growth is a consequence, not an input.



Four sectors carry three-quarters: office assistants/agents, advertising uplift, consumer, coding.

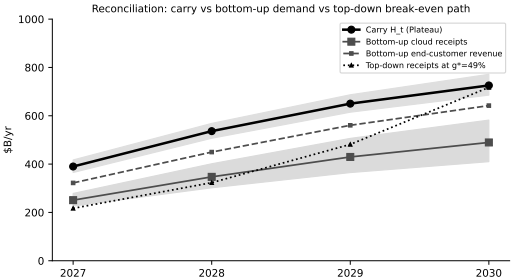
Ceilings: 350M seats $\times$ adoption $\times$ price, capped by IT budgets; Meta+Google ad yield lift; 1.3–2B users $\times$ paid conversion; 29M developers.

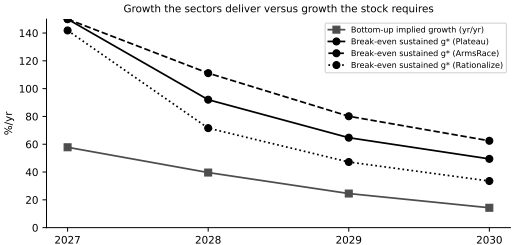
Implied growth: 57% (2027) $\rightarrow$ 39% $\rightarrow$ 24% $\rightarrow$ 14% (2030). CAGR 33%.

Source: paper Section 7, Table 4, Appendix B

Reconciliation: what the stock needs versus what the sectors deliver

The stock needs 48%/yr; the sectors deliver 33%. The 2030 gap — ~$255B of receipts — is 3.5 standard deviations of the propagated uncertainty.

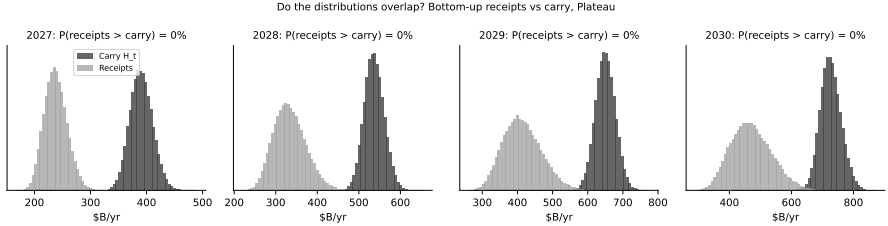




Source: paper Figs. 13–14; the two estimates share only the 2026 base

Do the distributions overlap?

Under a plateau or the street's path, receipts and carry do not touch in any year to 2030. Under a 35% cut they overlap from 2029.

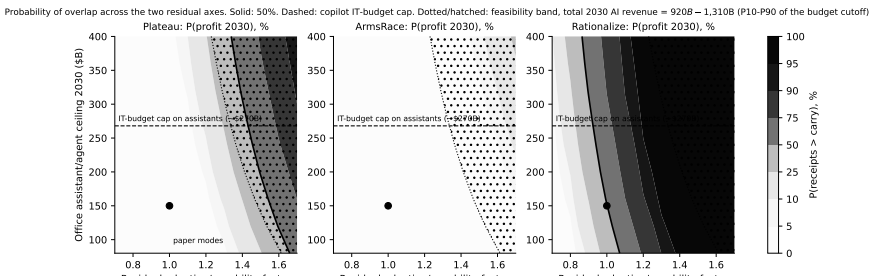


P(profit)	2027	2028	2029	2030
Plateau	0%	0%	0%	0%
Arms race	0%	0%	0%	0%
Rationalize	0%	2%	20%	52%

Source: paper Fig. 16, Table 13

The result as a probability map

Two residual axes — adoption beyond what the sectors see, and the assistant ceiling — inside a feasibility band set by enterprise budgets.

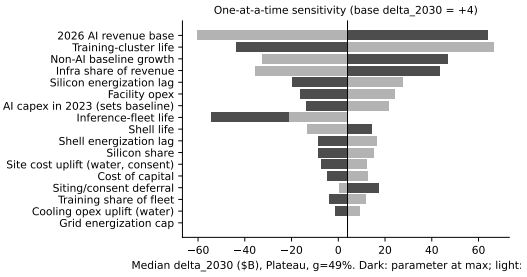


On a plateau the 50% contour sits at a residual factor ∼1.5 — inside the band, but requiring AI to take ∼25% of a software-and-services pool that is no longer growing. Under a cut, P(profit) is 52% in 2030 and 95% by 2032.

Source: paper Fig. 17

Where the uncertainty lives — and how decomposition moved it

Single judgment ranges: two inputs carried 82% of the variance. Decomposed into sourced components: root-sum-square 286 → 104.



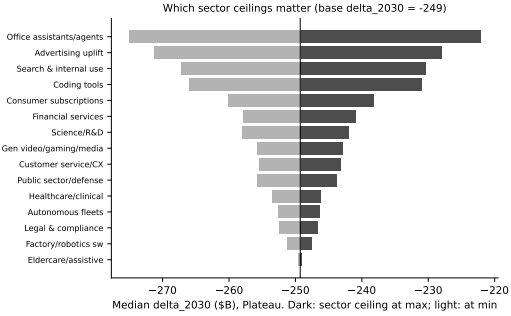
Input	before	after
2026 revenue base	64%	38%
Cloud share θ	18%	25%
Silicon life	5%	22%
Facility opex	7%	4%
Cost of capital	1%	8%
Grid, siting, lags	—	∼2%

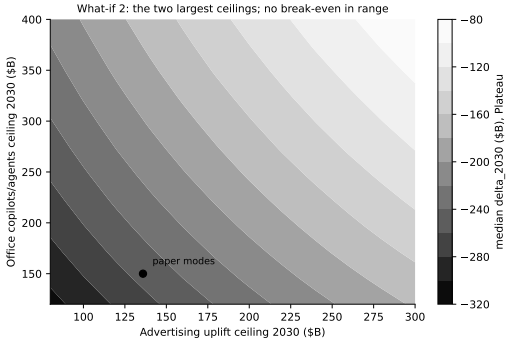
Decomposition tightened some inputs and moved others — the base down ($210B→$187B), opex down, the baseline up — in both directions, as an honest method should.

Source: paper Table 12, Section 8

No single sector closes the gap

Advertising and office assistants matter most; even both at their maxima leave the plateau short, because ramps and capture shares bind too.

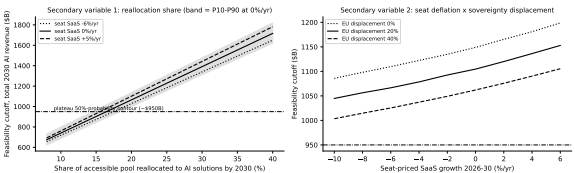




Source: paper Figs. 15 and 19

The budget the bull case draws on is itself being torn up

Seat-priced software deflating ($2T of market value gone Jan–Apr 2026), Europe leaving US vendors, services shrinking — yet the AI-financeable total is nearly unchanged, because the money moves from seats to solutions.



2030 software+services pool: $3.75T (not $4.85T at Gartner's growth); accessible to US vendors $3.25T; AI share ∼25%; band $920B–$1.31T.

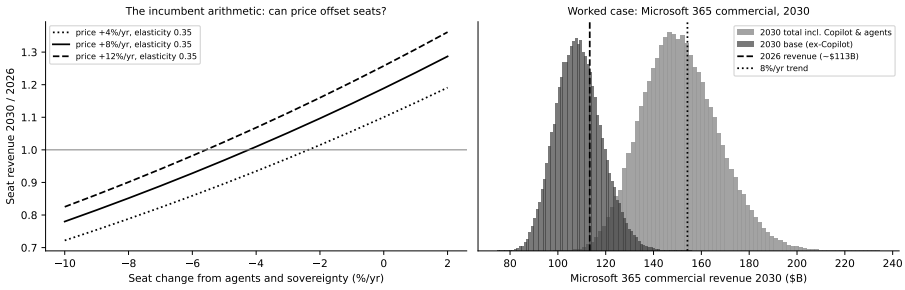
Composition changes far more than the total: incumbents lose the dollar, non-US providers keep more at home, the infrastructure meter sees a fraction.

Two terms are too loose to range and are swept instead: the reallocation share and seat-software growth.

Source: paper Section 8; Gartner; SaaSpocalypse coverage; State of Digital Sovereignty 2026

The incumbent's dilemma: adding AI while raising the bill

Revenue is seats × price. Price rises offset seat loss only while $(1+p)^4[1-\epsilon((1+p)^4-1)]$ beats the seat trend. Attach, not price, decides it.

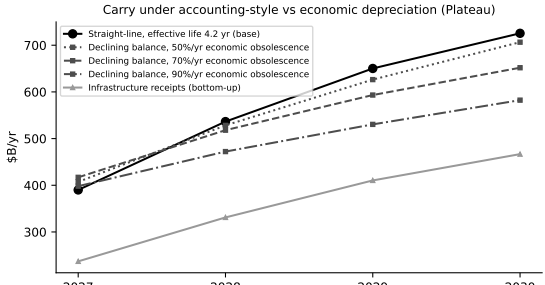


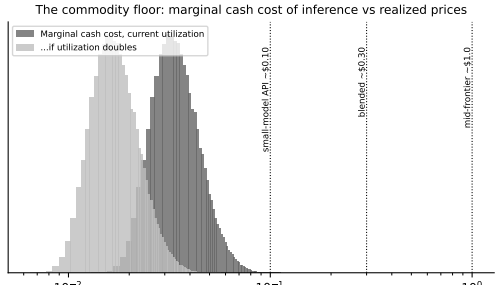
Microsoft as illustration, on unverified inputs: commercial seats still growing 6–7%/yr (8-K), Copilot paid seats 30M (6.5% attach; bundled via E7; usage 20–30% of paid in some estimates). The seat-deflation argument has not yet overcome the disclosed growth.

Source: paper Section 8; Microsoft 8-K FY27 metrics

Economic depreciation and the commodity floor

If an accelerator is worth what it earns against the newest chip, the 2023–26 silicon is already $100–250B below book. If token prices converge on marginal cash cost ($0.03–0.05/M), receipts converge on the opex line.



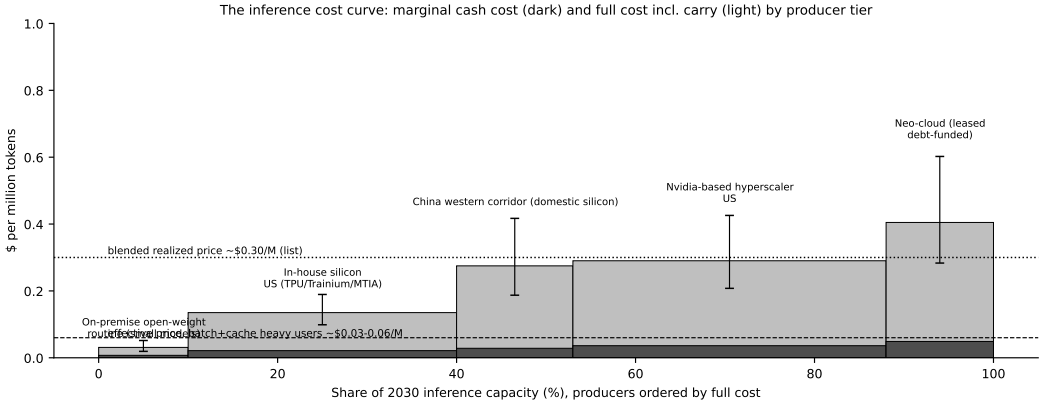


Counter-evidence, incorporated: H100 rentals *rising* to $2.63/hr in Sept 2026 (Silicon Data) — scarcity before the 2027–28 cohorts land. The variable stays swept until that series moves.

Source: paper Section 14; reviewer V. Mehta; Epoch AI; Silicon Data via a16z

Who is the low-cost producer

At effective prices, only in-house silicon on cheap power and on-premise open-weight routine work cover their carry. The debt-funded neo-cloud is the marginal producer the floor eliminates first.



Source: paper Fig. 25; EIA tariffs; Nvidia; Damodaran; Oracle utilization

Four decades of computing already taught this

Utilization gains go to the buyer, not the producer.

Mainframe batch, 1970s. A machine paid for by the shift is economic only if the night is filled with batch; operators ran at 90%+.

Bitcoin mining. Special-purpose silicon; within two years of each ASIC generation the hashprice converged to the marginal cost of the cheapest power; equipment obsolete in 12–24 months; capacity migrated to cheap electricity.

Inference has every one of those properties. Batch APIs at 50–75% off and caching at 90% off are the night shift arriving.

The productivity lag is measured, not asserted.

US labor productivity growth	%/yr
1948–73	2.8
1973–95 (PC era; Solow's paradox)	1.4
1995–2004 (IT payoff)	2.9
2005–19	1.4

Dynamo: ∼40 years (David 1990). PC: ∼15 years (Jorgenson–Ho–Stiroh 2008). Firm-level IT returns: 5–7 years, only with reorganization (Brynjolfsson–Hitt 2000).

Today: AI in production at ∼1 in 10 firms (Census BTOS 2025); 95% of pilots with no P&L impact (MIT NANDA 2025).

Source: paper Sections 9 and 14

Readiness: grid, water, consent

A funded, grid-connected project must also be allowed to be built.

Consent. $156B of US projects blocked or delayed in 2025; $130B in Q1 2026 alone (75 projects); 833 opposition groups in 49 states; New York moratorium July 2026; Ipsos ∼70% local opposition.

Water. A 2030 fleet of 60–110 GW consumes ∼90B gallons/yr evaporatively — under 0.1% of US withdrawals, decisive locally (Arizona, Aragon, the Randstad).

Rates. Data-center load *lowered* residential rates ∼6% over 2019–24 (Watten et al.); opposition has risen regardless.

In the model: siting deferral 10–35% of each cohort a year; shell cost uplift 2–9%; cooling opex uplift 0–6%; grid energization cap $1.15–1.68T/yr (binds only in arms-race draws).

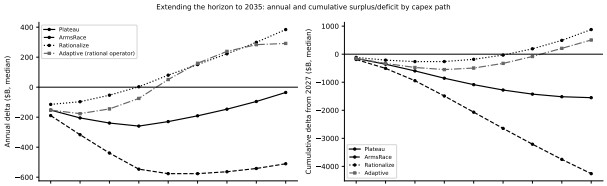
Effect: the carry shifts later, not away — 2027 down ∼$20B, 2030 up ∼$15B.

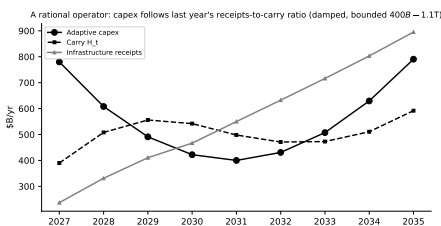
Sovereignty. France's national Linux mandate, Schleswig-Holstein, Denmark; German IT leaders implementing European software 20%→40% in a year. In the model: a 3–8% displacement ramp on sovereign-exposed sectors; κ=1.2, γ=0.25 overlay roughly doubles the 2030 deficit.

Source: paper Sections 8 and 10; Data Center Watch; Forbes; State of Digital Sovereignty 2026

Beyond 2030: a plateau does not recover by 2035

Receipts grow from $480B to $910B — but so does the carry, through replacement silicon and shell cohorts. A rational operator who cuts in step with receipts is profitable by 2031 and recovers cumulatively by 2034–35.

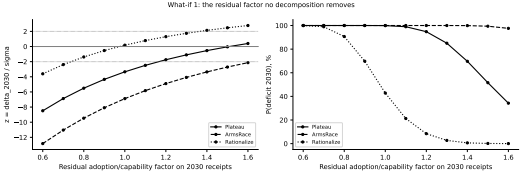


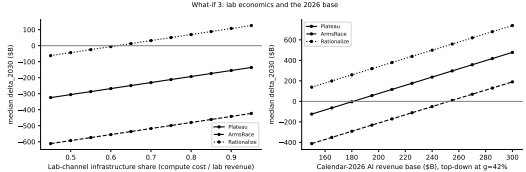


Source: paper Appendix E

What-if curves over the variables no decomposition removes

The plateau crosses $z = 0$ only if the sector chains are $\sim$50% too conservative at once.





Source: paper Figs. 18 and 20

The bull case at its strongest (a16z, September 2026) — where we agree

On nearly every observable, the two documents agree.

a16z's point	What the register does with it
The street keeps underestimating capex; consensus $1.05T (2027), $1.15T (2028)	Adopted: the paper's arms-race path *is* consensus; the plateau is below it. On consensus the 2030 deficit is ∼$545B
Backlogs doubled to ∼$1.7T	Observable added. A contract is not a payment: Oracle's converts over years, half from one counterparty
Hyperscaler ROIC ∼20% > cost of debt	True of the existing businesses; the AI cohorts' return is what the paper computes
Data centers lowered residential rates 6% (Watten et al.)	Taken as stated; the siting variable is calibrated on what communities do, not what rates did
Frontier labs: 69% of spend on 35% of tokens; tokens +600×	The cloud-capture share and the floor, measured from the demand side
SaaS: prove-it, not pocalypse; fundamentals stable	Why seat-software growth is a secondary variable; a16z's own newspaper chart is the caution

Source: paper Section 15

The bull case at its strongest — where we disagree, on two judgments

Not on the numbers: on who pays, and on whether rising compute prices are permanent.

1. Labor budgets or IT budgets?

a16z: AI's market is "considerably larger than IT budgets" because AI is paid from payroll — $30T, not $3.3T.

Paper: only about a quarter of nominal labor savings reach cash, over a J-curve (slack, verification, Parkinson–Jevons, the manager's dilemma). So the band is an IT-budget band.

If a16z is right, the band is wrong by a factor of several and the plateau pays. If the slack trap is right, the band holds. The register cannot settle this from 2026 data.

2. Scarcity or cycle top?

a16z: H100 rentals rising to $2.63/hr while token prices fall — Jevons absorbing every efficiency gain; GPUs "getting more valuable over time."

Paper: that is what Jevons looks like while supply is short. Bitcoin's precedent is what it looks like after supply catches up — and $2.63 against a $0.20–0.35 cash cost is the same 8–13× ratio the commodity floor finds at the token level.

The variable to watch: the H100 rental through 2027. A decline toward $1 is the floor arriving.

Source: paper Section 15

What would change the distribution

The result is a distribution, so the tests are observations that move it decisively.

Two hard tests, either sufficient:

1. Calendar-2028 AI revenue above \$400B on the paper's basis — i.e. $\geq$50%/yr has held two more years. Interim: 2027 near \$330B is our path; above \$350B favours the build-out.
2. By 2030: impairments under \$100B *and* no 30% capex retreat.

Softer, non-deciding: BLS displacement $<$1.5%; frontier API prices $-$80% with integrator revenue growing.

The calendar

Oct 28–30	Big-4 Q3 calls: 2027 guidance, Copilo
Nov	Anthropic IPO: first audited lab figure
Nov 19	Nvidia Q3 FY27
by Dec 3	**v27 issued** with a delta log
Jan 2027	10-Ks: calendar-2026 revenue; useful
Jan 2028	Interim marker (\$330B / \$350B)
Jan 2029	The revenue test (\$400B)

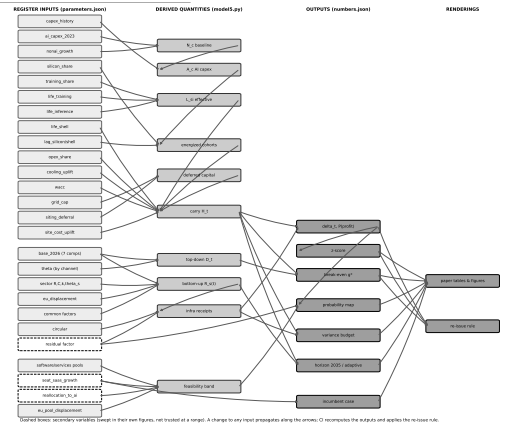
Source: paper Section 17 and Appendix H

Every input, appraised — and graded

Forty-plus register entries, each with evidence, range, the likely bias of its source, and what would improve it. Graded like Wikipedia's article classes.

Grade	Meaning
P	anchored on a primary source
S	secondary, or chain with judgment steps
J	judgment, no external anchor — the first
—	secondary variables, swept not ranged

The companies concerned could retire most of the J and S entries by disclosing a few numbers they already have: paid seats and their trend, AI revenue on a stated basis, compute cost as a share of lab revenue, accelerator lives in service.



Source: paper Appendix G; community_repo/parameters.json; blueprint/

How this estimate is improved: a community model

One object of record, small sourced contributions, machinery that checks them, collective credit.

Borrowed from collaborative mathematics (Polymath; the Equational Theories Project) and from open policy models (Tax-Calculator, OpenFisca):

1. One GitHub issue per input: the talk page.
2. A pull request changing `parameters.json`: the edit.
3. CI rebuilds and posts the consequence (z, g^*, P) before the edit lands.
4. A reviewer checks the source, never the conclusion.
5. Versioned releases with a Zenodo DOI, a colour-marked change copy, and a delta log.

The rules. Provenance or it did not happen. Ranges, not points. Say who benefits. No result-driven edits. Two defensible ranges for one input ⇒ a secondary variable until a disclosure settles it.

Re-issue rule. Minor if $|\Delta z| < 0.5$, $|\Delta g^*| < 5$ pts, $|\Delta P| < 15$ pts; material otherwise, or if an observable lands outside its P10–P90.

Best first contribution: a primary source for a number we could only infer.

Source: community_repo: README, CONTRIBUTING, GOVERNANCE, COMMUNITY_REVIEW

The whole thing, one slide

- The carry of the installed base ramps from ∼$390B (2027) to ∼$725B (2030) on a plateau; the street's path (arms race) takes it past $1T.
- 2027–28 are a sunk deficit under every path.
- Break-even needs ∼48%/yr sustained to 2030 on a plateau, ∼62% on consensus; the sector chains deliver ∼33%, decaying to 14%.
- The 2030 shortfall on a plateau is ∼$255B, 3.5$\sigma$; P(profit) is zero on a plateau and on consensus, 52% under a 35% cut, 95% by 2032.
- AI revenue can approach $1T while software shrinks, because most of it is the software industry's own money changing hands — and only a fraction reaches the meter.
- If token prices find their marginal cash cost, the carry is never recovered by anyone but the lowest-cost producer: in-house silicon on cheap power running open-weight models.
- A plateau does not recover by 2035; a rational operator who cuts in step with receipts is profitable by 2031.
- The two judgments the result turns on: whether AI is paid from labor budgets, and whether today's rising GPU rentals are permanent. The register names both; the October calls begin to answer them.

Paper, register, model, figures: Zenodo DOI (on deposit) and the public repository. Reviewers: Francis Pedraza, Matthew Fitzpatrick, Viraj Mehta.